EE708 Course Project

ShiftGuard-10

Robust Image Classification Challenge

Group 20

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FINAL F1 SCORE

88.3%

Problem Statement

Task

10-class image classification on 32×32 RGB images. We are given 29,400 training images and 7,600 test images across 10 classes.

Challenge

Test images come from a shifted distribution. Models that memorise training patterns will fail at test time.

Metric

Macro F1 Score — every class is weighted equally regardless of sample count, making it robust to severe class imbalance.

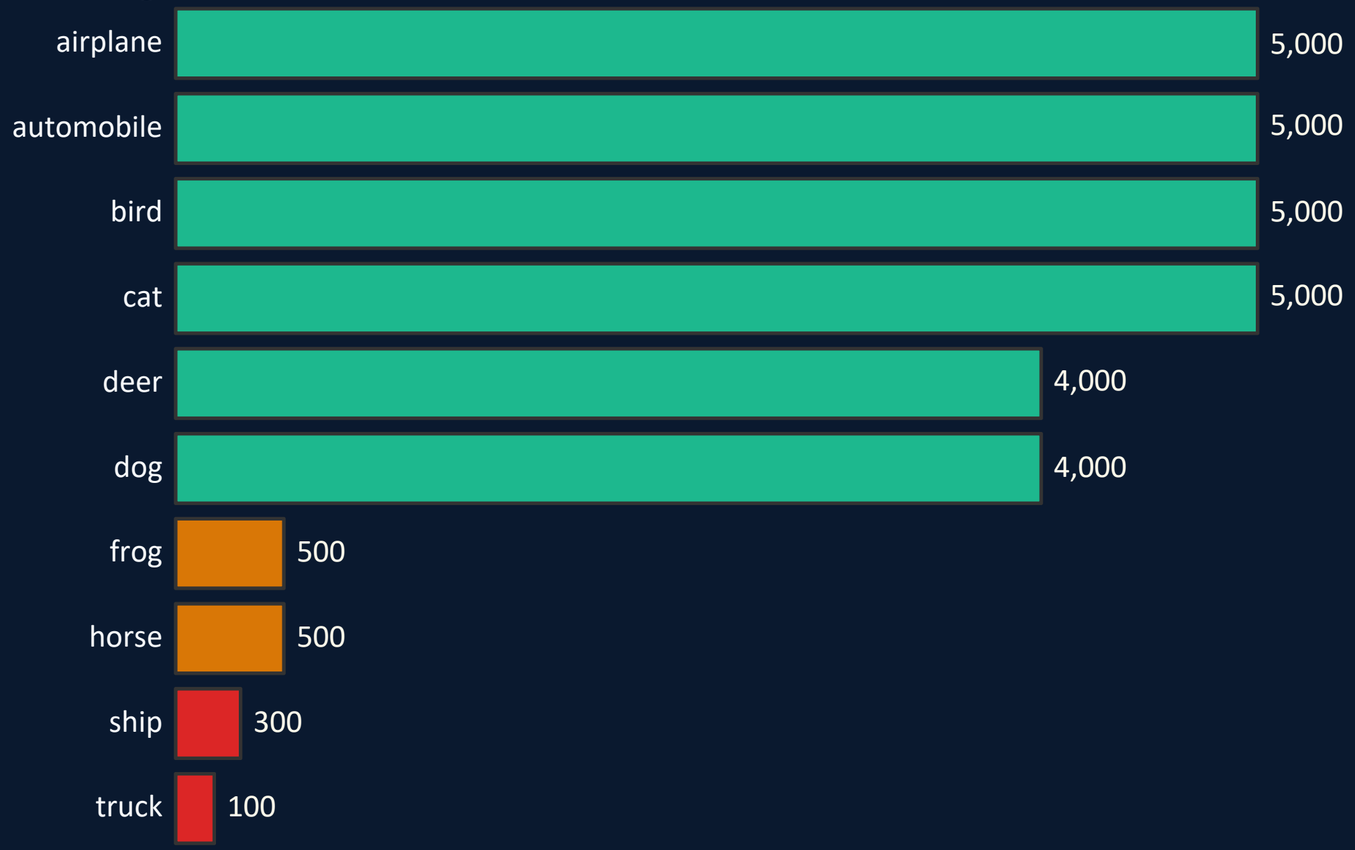
Goal

Learn shift-invariant, robust features that consistently generalise beyond the initial training distribution.

Dataset Understanding / EDA

29,400	7,600	10	32×32	90 / 10
Training images	Test images	Classes	Image size (RGB)	Train / Val split

Training Class Distribution



Severe Class Imbalance

Airplane, automobile, bird, cat → 5,000 samples each
Deer, dog → 4,000 samples
Frog, horse → 500 | Ship → 300 | Truck → only 100!

How We Handled Imbalance

1. **Macro F1 metric** equally weights all 10 classes
2. **Aggressive augmentation** — each of the 100 truck images appears with different augmentations every epoch
3. **Mixup / CutMix** creates synthetic blended samples, amplifying rare classes

Baseline to Final Model — Journey

1 **Early Attempts**

Basic CNN, standard augmentation



67.3%

2 **Better Augmentation**

Improved data augmentation pipeline



72.1%

3 **WRN-28-6 (single)**

Wide ResNet + RandAugment + Mixup/CutMix



87.84%

4 **Ensemble × 2 + TTA**

Two diverse WRN-28-6 models, 15-pass TTA



88.34%

Model Architecture — WideResNet-28-6

Depth = 28, Widen Factor = 6

4× wider channels than standard ResNet-28.

Filter progression

16 → 96 → 192 → 384 across 3 block groups (×4 blocks each)

Pre-activation residual block

BN → ReLU → Conv3×3 → Dropout → BN → ReLU → Conv3×3

Dropout

0.30 (Model-1) / 0.25 (Model-2) — different per model for ensemble diversity

Weight decay

5×10^{-4} — L2 regularisation on all conv layers

Global Average Pooling

Replaces flatten + large FC layer; reduces overfitting



Augmentation Strategy

01 RandAugment

What: 13 candidate ops (rotate, shear, solarize, equalize, brightness, contrast, sharpness…). 2 randomly chosen per image at magnitude 9/30.

Why: Simulates photometric & geometric distribution shifts — the primary challenge of this competition.

02 Spatial Augmentation

What: Random horizontal flip (50%) + reflect-padded random crop (±4px).

Why: Forces position-invariant features; padding with reflect mode avoids black-border artifacts.

03 Cutout (×2 holes)

What: Two 8×8 pixel regions zeroed out per image.

Why: Forces model to use global context, not isolated textures — critical for robustness to occlusion.

04 Mixup / CutMix (50/50)

What: α=0.2 Mixup or α=1.0 CutMix randomly per batch. Produces soft labels.

Why: Reduces overconfidence; amplifies rare classes like truck (only 100 samples) through blended samples.

Training Configuration

Parameter	Value
Optimiser	SGD + Nesterov momentum (0.9)
Loss	CategoricalCE + Label Smoothing 0.05
Epochs	200
Batch Size	128
Initial LR	0.1
LR Schedule	5-epoch linear warmup → cosine annealing
Model 1	Seed 42, Dropout 0.30
Model 2	Seed 123, Dropout 0.25
Checkpoint	Best val_accuracy saved per model

LR Schedule

Warmup (ep 1–5)

LR grows linearly from 0 → 0.1 over 5 epochs. Prevents early instability with large LR before model is partially trained.

Cosine Annealing (ep 6–200)

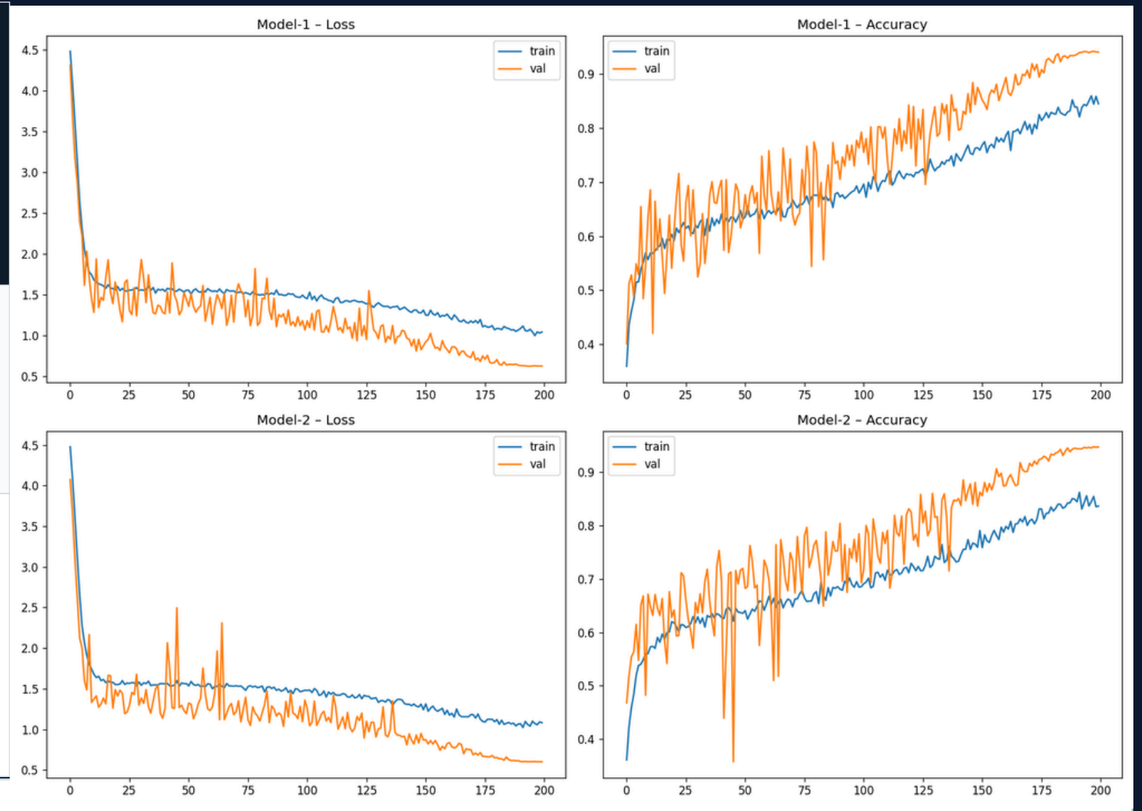
LR = 0.1 × 0.5 × (1 + cos(π × progress)). Smooth decay from 0.1 → ~0. Explores broadly then fine-tunes precisely.

Why not step decay?

Step decay drops LR abruptly, missing good minima between steps. Cosine annealing converges more stably to lower loss.

Training Curves — Both Models

Model	Train Acc (ep 200)	Val Acc (ep 200)	Best Val Acc
Model-1 (seed=42, drop=0.30)	84.5%	94.05%	**94.18%**
Model-2 (seed=123, drop=0.25)	83.6%	94.7%	**94.73%**



Val accuracy consistently above train accuracy

This is expected and healthy — NOT overfitting. Heavy augmentation (Mixup, CutMix, Cutout, RandAugment) makes the training objective much harder than the clean val evaluation. Train accuracy is computed on blended/occluded images; val on clean images.

Training loss stays elevated

Because Mixup/CutMix produce soft labels, the model can never achieve zero loss. This is by design — it prevents overconfidence and improves calibration on the shifted test distribution.

Ensemble + TTA



- Probability averaging (not voting) — uses full confidence information from softmax outputs
- 30 effective prediction paths: 2 models × 15 TTA passes each
- Different seeds per model ensure uncorrelated augmentation sequences and weight initializations

Results & Key Takeaways

88.3%	90.90%	89.87%	92.48%
Test F1 — Competition	Ensemble Val F1 (with TTA)	Model-1 Val F1 (with TTA)	Model-2 Val F1 (with TTA)
Final score	Average of both models	seed=42, drop=0.30	seed=123, drop=0.25

Note: All individual model F1s above are also computed with 15-pass TTA. The simple average ensemble gives equal weight to both models — a weighted ensemble favouring Model-2 (stronger on clean val) was not needed as the ensemble provides more robust predictions on the shifted test distribution.

Submission class distribution (from code output)

Class	Predictions	Training Samples
ship	906	300
frog	905	500
dog	855	4000
deer	851	4000
horse	833	500
automobile	722	5000
truck	**698**	**100**
airplane	636	5000
bird	599	5000
cat	595	5000

Key Takeaways

- Augmentation diversity is the #1 weapon against distribution shift
- WRN width > depth for 32×32 images — wider features generalise better
- Ensemble diversity (seed + dropout) gives consistent +1% gain over a single model
- Truck (100 training samples) received 698 predictions — augmentation solved the imbalance
- Probability averaging + TTA is computationally cheap and almost always worth it